K.R. MANGALAM UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

LAB FILE

COURSE NAME: COMPUTER NETWORK

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COURSE: BTECH CSE (AI and ML)

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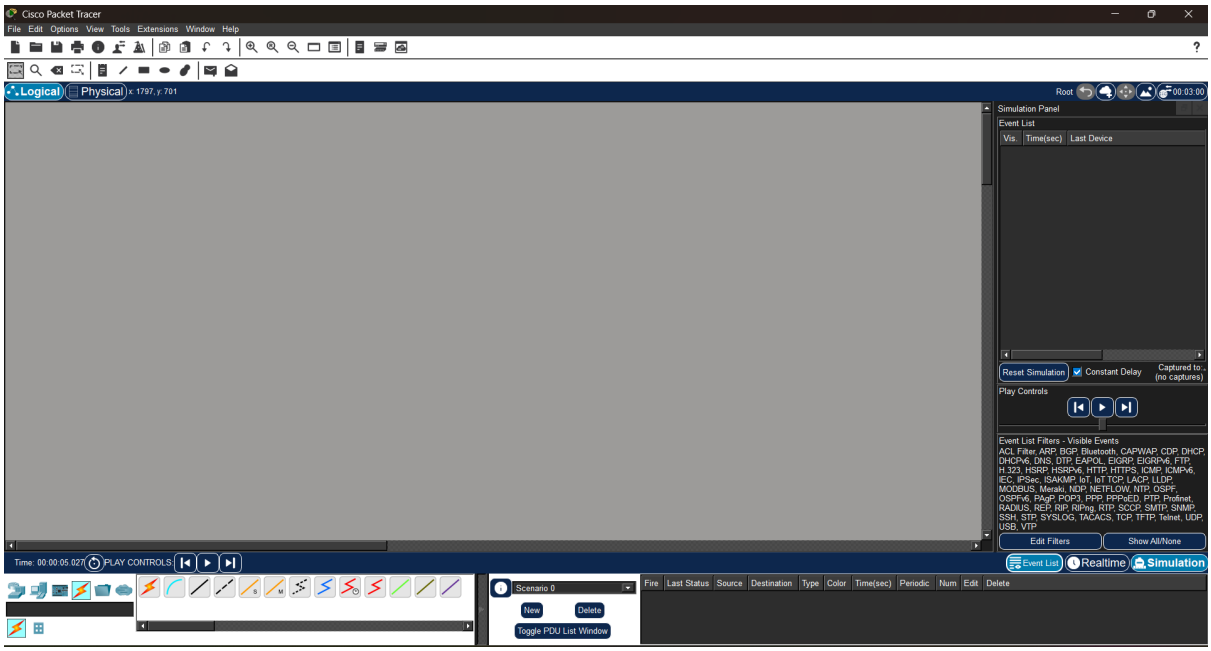
EXPERIMENT 1

INTRODUCTION TO COMPUTER NETWORKS AND THE CISCO PACKET TRACER (CPT)

A. CPT Platform Overview



Key Features of Cisco Packet Tracer

1. Workspace

The workspace is the main area where network topologies are created and simulations are performed. It allows users to design networks, monitor packet flow, and analyze performance.

2. Menu Bar

The menu bar provides access to various options such as File, Edit, Options, View, Tools, Extensions, Window, and Help. These options are used to manage files and configure the application.

3. Main Toolbar

The main toolbar contains shortcut icons for commonly used functions like saving, opening files, and zooming, making tasks quicker and more efficient.

4. Common Tools Bar

This bar includes frequently used tools such as Select, Inspect, Delete, Resize Shape, Place Note, Drawing Palette, and PDU tools for creating and managing network interactions.

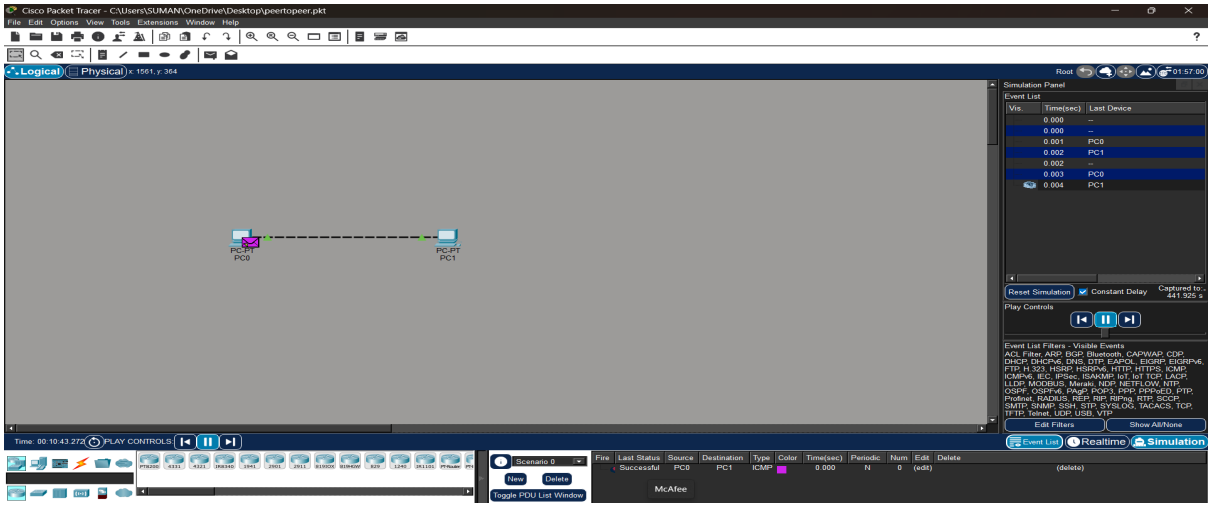
5. Logical/Physical Workspace and Navigation Bar
This feature allows switching between Logical Workspace (network design view) and Physical Workspace (real-world layout view), helping users visualize networks effectively.

6. Device-Type Selection Box
This box displays different categories of devices such as routers, switches, and end devices. Selecting a category shows the specific devices available for use.

7. User-Created Packet Window
This window is used in simulation mode to manage and observe packets created by the user, helping in analyzing network behavior.

8. Realtime/Simulation Bar
This bar allows users to switch between Realtime and Simulation modes. It also provides controls like play, pause, and fast forward to observe packet transmission step by step.

B. Peer to peer network



<u>Peer-to-Peer Network Implementation</u>

Aim:

To create and configure a peer-to-peer network between two computers and verify communication.

Requirements:

- 2 PCs
- 1 Copper Cross-Over Cable

Procedure :

1. Open Cisco Packet Tracer.
2. From the End Devices section, drag and place 2 PCs (PC1 and PC2) onto the workspace.
3. Select Connections and choose Copper Cross-Over Cable.
4. Connect:
    - PC1 → FastEthernet0
    - PC2 → FastEthernet0
5. Configure IP Address on PC1:
    - Go to Desktop → IP Configuration
    - IP Address: 192.168.1.1

    - Subnet Mask: 255.255.255.0
6. Configure IP Address on PC2:
    - Go to Desktop → IP Configuration
    - IP Address: 192.168.1.2
    - Subnet Mask: 255.255.255.0
7. Verify connectivity:
    - Open Command Prompt on PC1
    - Type: ping 192.168.1.2

IP Configuration:

PC1: 192.168.1.1 / 255.255.255.0
PC2: 192.168.1.2 / 255.255.255.0

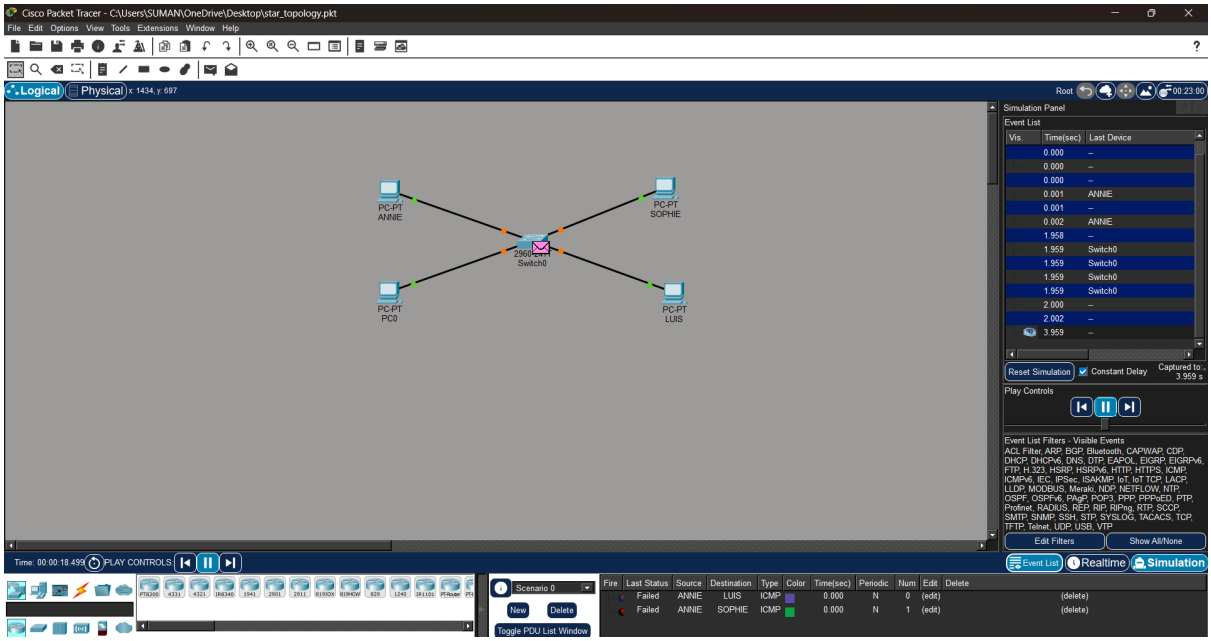
Result:

Both PCs successfully communicated with each other using a peer-to-peer connection. The ping test was successful, indicating proper network configuration and direct communication without any central server.

EXPERIMENT2

Network Topologies

A. Star Topology



Star Topology Implementation

Aim:

To implement a Star Topology network using a switch and verify communication between devices.

Requirements:

- 1 Switch (2960)
- 4 PCs (ANNIE, SOPHIE, LUIS, PC0)
- Copper Straight-Through Cables

Procedure:

1. Open **Cisco Packet Tracer**.
2. Add devices to workspace:
    - One **2960 Switch**

    - Four PCs named **ANNIE, SOPHIE, LUIS, PC0**
3. Connect each PC to the switch using **Copper Straight-Through Cable**:
    - ANNIE → Switch
    - SOPHIE → Switch
    - LUIS → Switch
    - PC0 → Switch
4. Configure IP Addresses:
    - **ANNIE:** 192.168.1.1 / 255.255.255.0
    - **SOPHIE:** 192.168.1.2 / 255.255.255.0
    - **LUIS:** 192.168.1.3 / 255.255.255.0
    - **PC0:** 192.168.1.4 / 255.255.255.0
5. Verify connectivity using **Simulation Mode**:
    - Send Simple PDU or use ping command
    - Observe packet flow in Event List

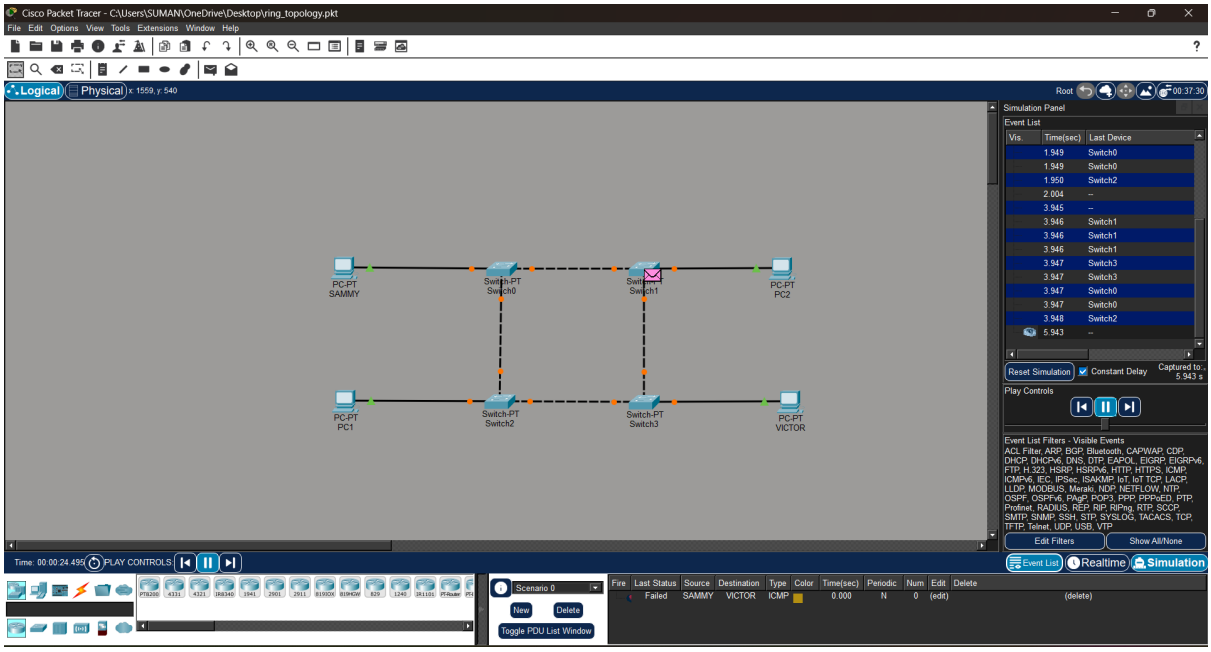
IP Configuration:

ANNIE: 192.168.1.1 / 255.255.255.0
SOPHIE: 192.168.1.2 / 255.255.255.0
LUIS: 192.168.1.3 / 255.255.255.0
PC0: 192.168.1.4 / 255.255.255.0

Result:

Initially, some packets may show **failed status** in simulation (as seen in your screenshot) due to ARP resolution. After a few seconds, communication becomes successful between all devices through the switch.

A. Ring Topology



Ring Topology Implementation

Aim:

To implement a Star Topology network using a switch and verify communication between devices.

Requirements:

- 1 Switch (2960)
- 4 PCs (ANNIE, SOPHIE, LUIS, PC0)
- Copper Straight-Through Cables

Procedure / Implementation Steps:

1. Open **Cisco Packet Tracer**.
2. Add devices to workspace:
    - One **2960 Switch**
    - Four PCs named **ANNIE, SOPHIE, LUIS, PC0**

3. Connect each PC to the switch using **Copper Straight-Through Cable**:
    - ANNIE → Switch
    - SOPHIE → Switch
    - LUIS → Switch
    - PC0 → Switch
4. Configure IP Addresses:
    - **ANNIE:** 192.168.1.1 / 255.255.255.0
    - **SOPHIE:** 192.168.1.2 / 255.255.255.0
    - **LUIS:** 192.168.1.3 / 255.255.255.0
    - **PC0:** 192.168.1.4 / 255.255.255.0
5. Verify connectivity using **Simulation Mode**:
    - Send Simple PDU or use ping command
    - Observe packet flow in Event List

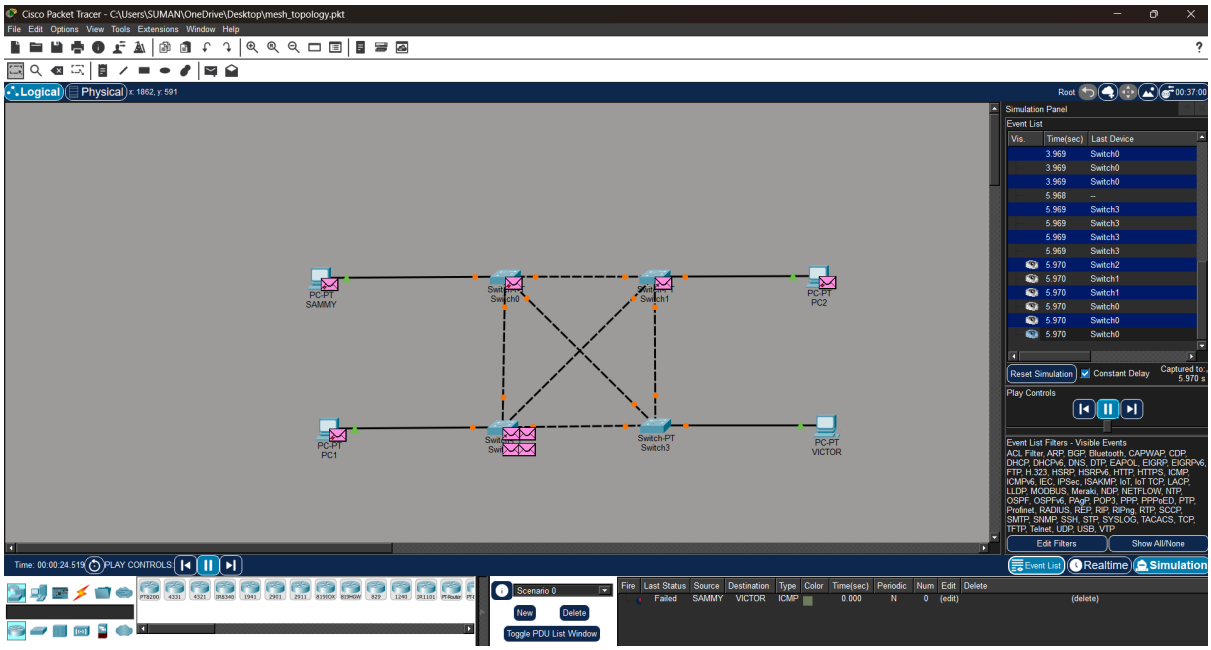
IP Configuration:

ANNIE: 192.168.1.1 / 255.255.255.0
SOPHIE: 192.168.1.2 / 255.255.255.0
LUIS: 192.168.1.3 / 255.255.255.0
PC0: 192.168.1.4 / 255.255.255.0

Result:

Initially, some packets may show **failed status** in simulation (as seen in your screenshot) due to ARP resolution. After a few seconds, communication becomes successful between all devices through the switch.

B. Mesh Topology



Mesh Topology Implementation

Aim:

To design and implement a Mesh Topology network and verify communication between devices.

Requirements:

- 4 Switches
- 4 PCs (SAMMY, PC1, PC2, VICTOR)
- Copper Straight-Through Cables

Procedure:

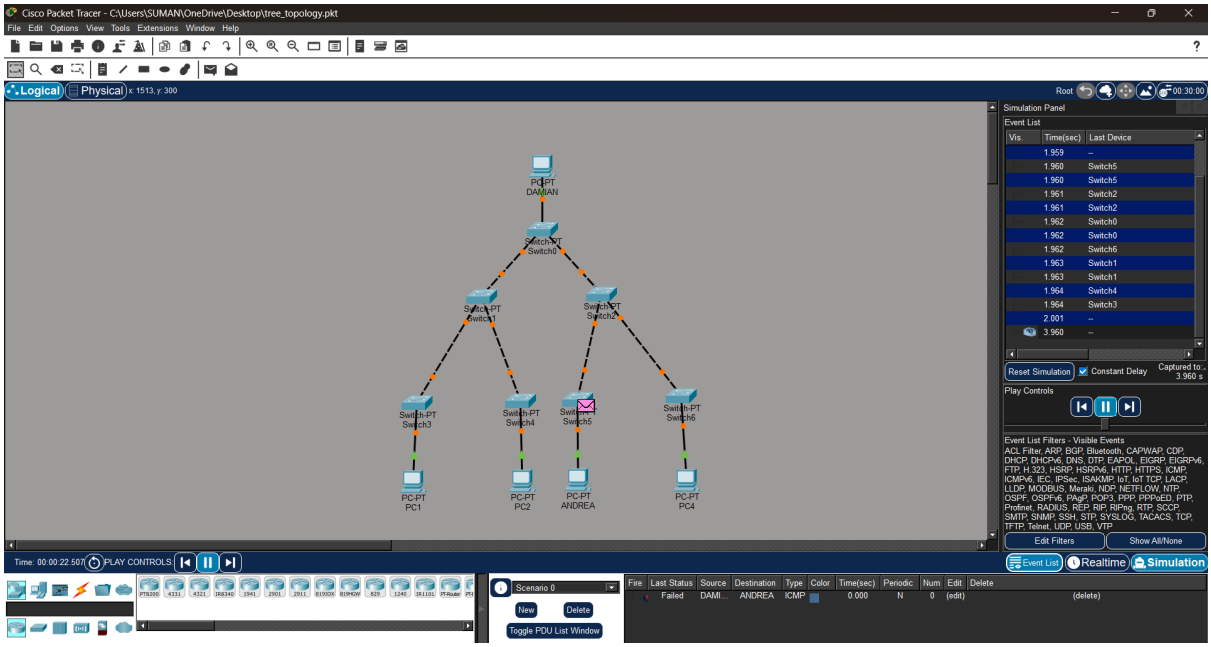
1. Open **Cisco Packet Tracer**.
2. Add devices to the workspace:
    - 4 switches (Switch0, Switch1, Switch2, Switch3)

    - 4 PCs (SAMMY, PC1, PC2, VICTOR)
3. Connect switches in **full mesh topology**:
    - Each switch is connected to every other switch
4. Connect PCs to switches:
    - SAMMY → Switch0
    - PC2 → Switch1
    - PC1 → Switch2
    - VICTOR → Switch3
5. Configure IP Addresses:
    - SAMMY: 192.168.1.1 / 255.255.255.0
    - PC1: 192.168.1.2 / 255.255.255.0
    - PC2: 192.168.1.3 / 255.255.255.0
    - VICTOR: 192.168.1.4 / 255.255.255.0
6. Verify connectivity:
    - Use ping command or simulation mode
    - Observe multiple paths for packet transmission

IP Configuration:
SAMMY: 192.168.1.1 / 255.255.255.0
PC1: 192.168.1.2 / 255.255.255.0
PC2: 192.168.1.3 / 255.255.255.0
VICTOR: 192.168.1.4 / 255.255.255.0

Result:
Initial packet failure may occur due to ARP process. After resolution, all devices successfully communicate using multiple redundant paths available in the mesh topology.

D. Tree Topology



Tree Topology Implementation

Aim:

To design and implement a Tree Topology network and verify communication between devices.

Requirements:

- Multiple Switches (Switch0 to Switch6)
- PCs (DAMIAN, PC1, PC2, ANDREA, PC4)
- Copper Straight-Through Cables

Procedure:

1. Open **Cisco Packet Tracer**.
2. Add devices to the workspace:
    - Several switches arranged in hierarchical structure
    - PCs connected at different levels

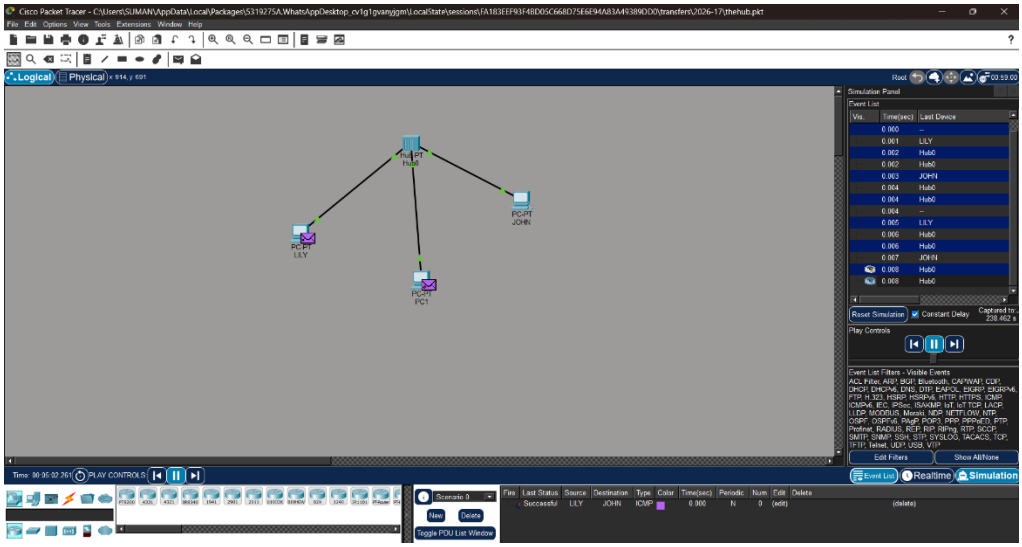
3. Create tree structure:
    - One **root switch (Switch0)** at top
    - Connect it to intermediate switches (Switch1, Switch2)
4. Connect PCs to leaf switches:
    - DAMIAN → Switch0
    - PC1 → Switch3
    - PC2 → Switch4
    - ANDREA → Switch5
    - PC4 → Switch6
5. Configure IP Addresses:
    - DAMIAN: 192.168.1.1 / 255.255.255.0
    - PC1: 192.168.1.2 / 255.255.255.0
    - PC2: 192.168.1.3 / 255.255.255.0
    - ANDREA: 192.168.1.4 / 255.255.255.0
    - PC4: 192.168.1.5 / 255.255.255.0
6. Verify connectivity:
    - Use ping command or simulation mode
    - Observe data flow from root to leaf nodes

IP Configuration:
DAMIAN: 192.168.1.1 / 255.255.255.0
PC1: 192.168.1.2 / 255.255.255.0
PC2: 192.168.1.3 / 255.255.255.0
ANDREA: 192.168.1.4 / 255.255.255.0
PC4: 192.168.1.5 / 255.255.255.0

Result:
Initial packet transmission may fail due to ARP resolution. After proper address resolution, all devices successfully communicate across the hierarchical network.

EXPERIMENT3

A. LAN using Hub



Network Implementation

Aim:

To create a Local Area Network (LAN) using a hub and verify communication between devices.

Requirements:

- 1 Hub
- 3 PCs (LILY, JOHN, PC1)
- Copper Straight-Through Cables

Procedure:

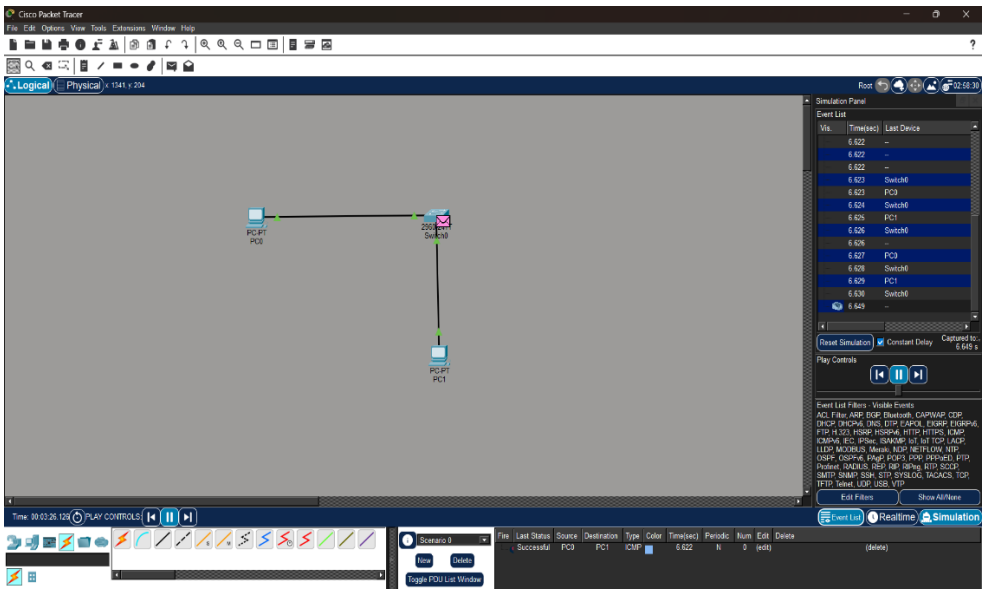
1. Open **Cisco Packet Tracer**.
2. Add devices to the workspace:
    - One **Hub (Hub0)**
    - Three PCs (**LILY, JOHN, PC1**)
3. Connect each PC to the hub using **Copper Straight-Through Cable**:
    - LILY → Hub

    - JOHN → Hub
    - PC1 → Hub
4. Configure IP Addresses:
    - LILY: 192.168.1.1 / 255.255.255.0
    - JOHN: 192.168.1.2 / 255.255.255.0
    - PC1: 192.168.1.3 / 255.255.255.0
5. Verify connectivity:
    - Use ping command or simulation mode
    - Observe data transmission via hub

IP Configuration:
LILY: 192.168.1.1 / 255.255.255.0
JOHN: 192.168.1.2 / 255.255.255.0
PC1: 192.168.1.3 / 255.255.255.0

Result:
All devices successfully communicated with each other. The hub broadcasted data to all connected devices, allowing communication within the LAN.

B. **LAN using switch**



Network Implementation

Aim:
To create a Local Area Network (LAN) using a switch and verify communication between devices.

Requirements:

- 1 Switch (2960)
- 2 PCs (PC0, PC1)
- Copper Straight-Through Cables

Procedure / Implementation Steps:

1. Open **Cisco Packet Tracer**.
2. Add devices to the workspace:
    - One **Switch (2960)**
    - Two PCs (**PC0, PC1**)
3. Connect PCs to the switch using **Copper Straight-Through Cable**:

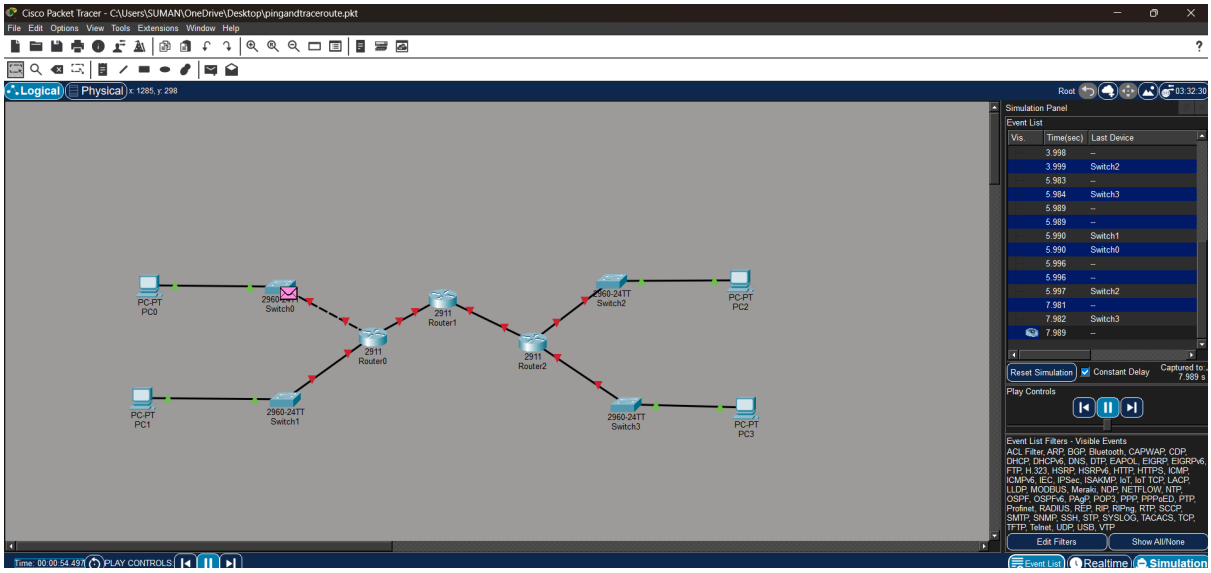
    - PC0 → Switch
    - PC1 → Switch
4. Configure IP Addresses:
    - PC0: 192.168.1.1 / 255.255.255.0
    - PC1: 192.168.1.2 / 255.255.255.0
5. Verify connectivity:
    - Use ping command or simulation mode
    - Observe successful packet transmission

IP Configuration:
PC0: 192.168.1.1 / 255.255.255.0
PC1: 192.168.1.2 / 255.255.255.0

Result:
Both devices successfully communicated with each other through the switch. The switch forwarded data only to the destination device, ensuring efficient communication.

EXPERIMENT4

Tracer and Ping commands-Network Diagnostic



Network Implementation

Aim:

To study and implement network diagnostic tools like ping and traceroute to test connectivity and analyze the path between devices.

Requirements:

- 3 Routers (Router0, Router1, Router2)
- 4 Switches
- 4 PCs (PC0, PC1, PC2, PC3)
- Copper Straight-Through and Serial Cables

Procedure / Implementation Steps:

1. Open **Cisco Packet Tracer**.
2. Create network topology:
    - Connect PCs to switches
    - Connect switches to routers

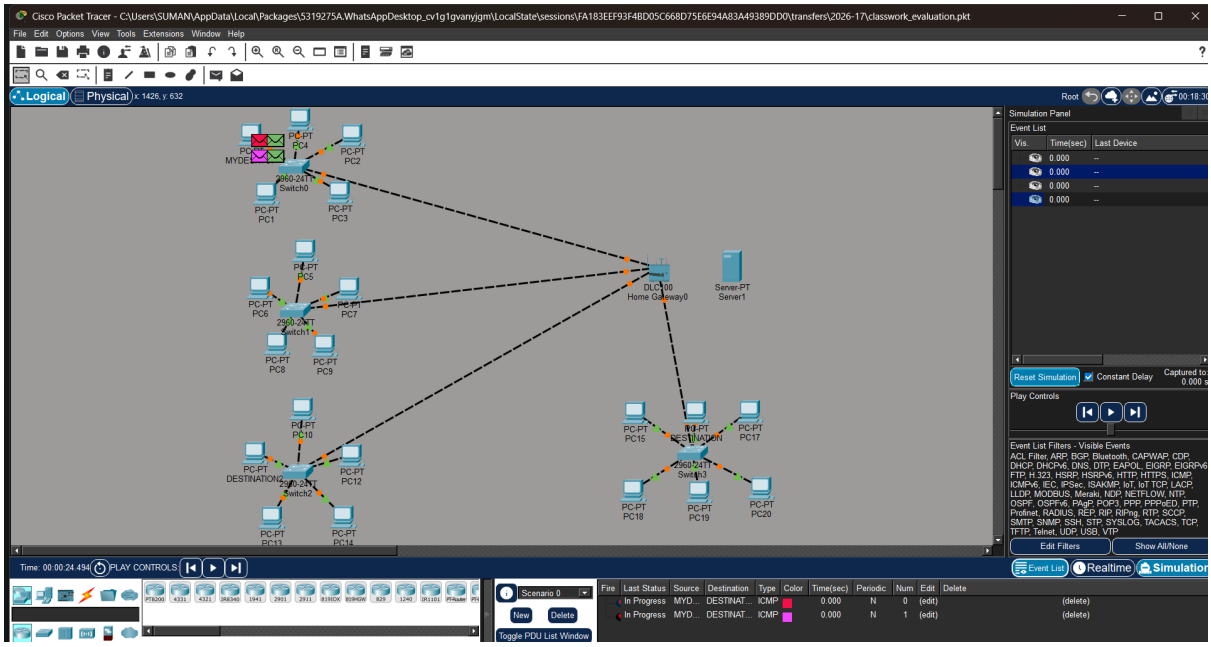
    - Connect routers to each other (forming a path)
3. Configure IP Addresses on PCs and interfaces of routers.
4. Configure routing (Static or Dynamic) to ensure end-to-end connectivity.
5. Test using **Ping command**:
    - Open Command Prompt on PC0
    - Type: ping <destination IP>
    - Example: ping PC2 IP
6. Test using **Traceroute command**:
    - Type: tracert <destination IP>
    - It shows the path (routers) taken by packets

IP Configuration (Example):
PC0: 192.168.1.1
PC1: 192.168.2.1
PC2: 192.168.3.1
PC3: 192.168.4.1

Result:
Ping command successfully verified connectivity between devices. Traceroute displayed the path taken by packets through multiple routers, showing each hop in the network.

EXPERIMENT5

Design a campus area network



Network Implementation

Aim:

To design and implement a Campus Area Network (CAN) connecting multiple departments/buildings and verify communication.

Requirements:

- Multiple Switches (2960 series)
- Multiple PCs (PC0 to PC20)
- 1 Home Gateway (central device)
- 1 Server
- Copper Straight-Through Cables

Procedure / Implementation Steps:

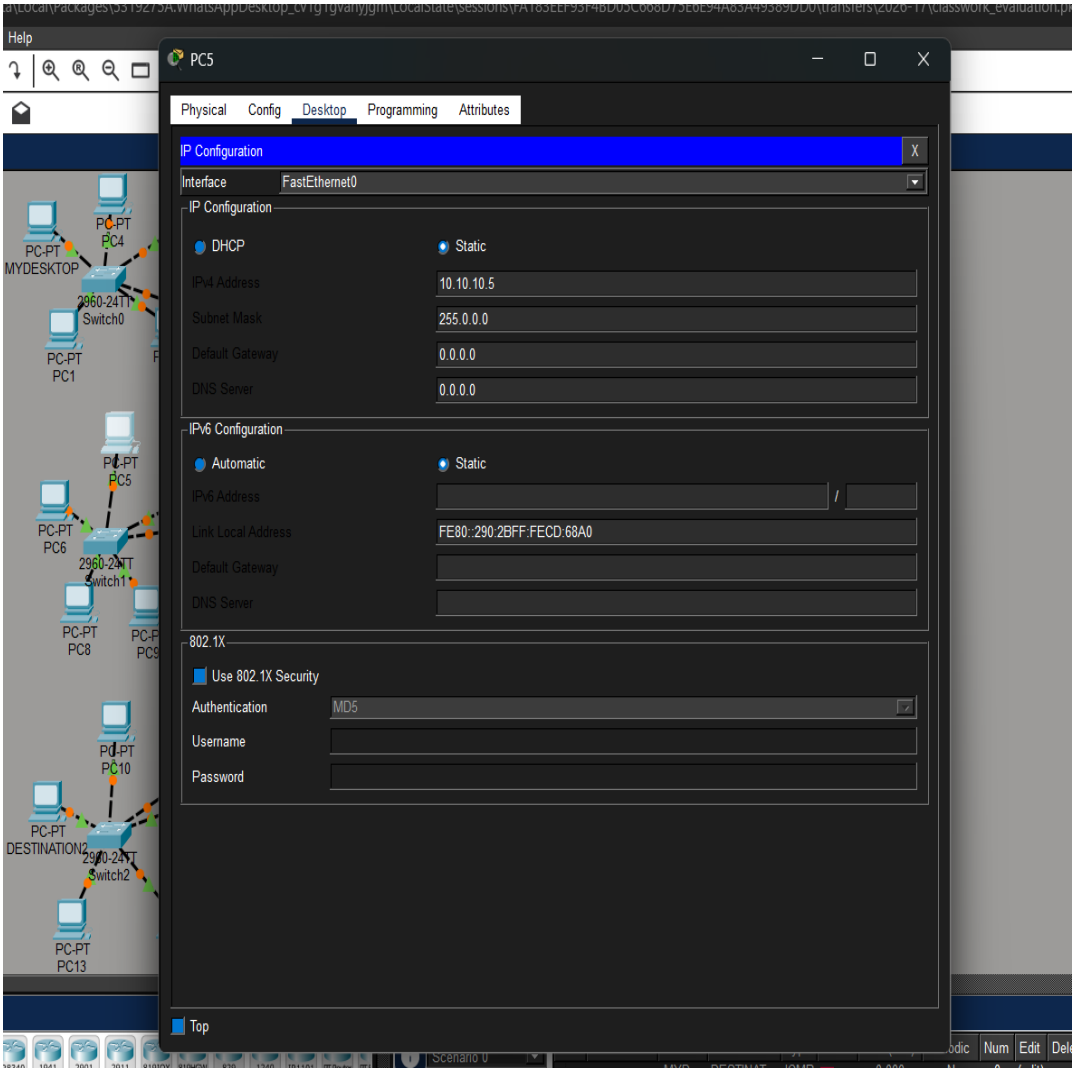
1. Open **Cisco Packet Tracer**.
2. Create different network segments (departments):

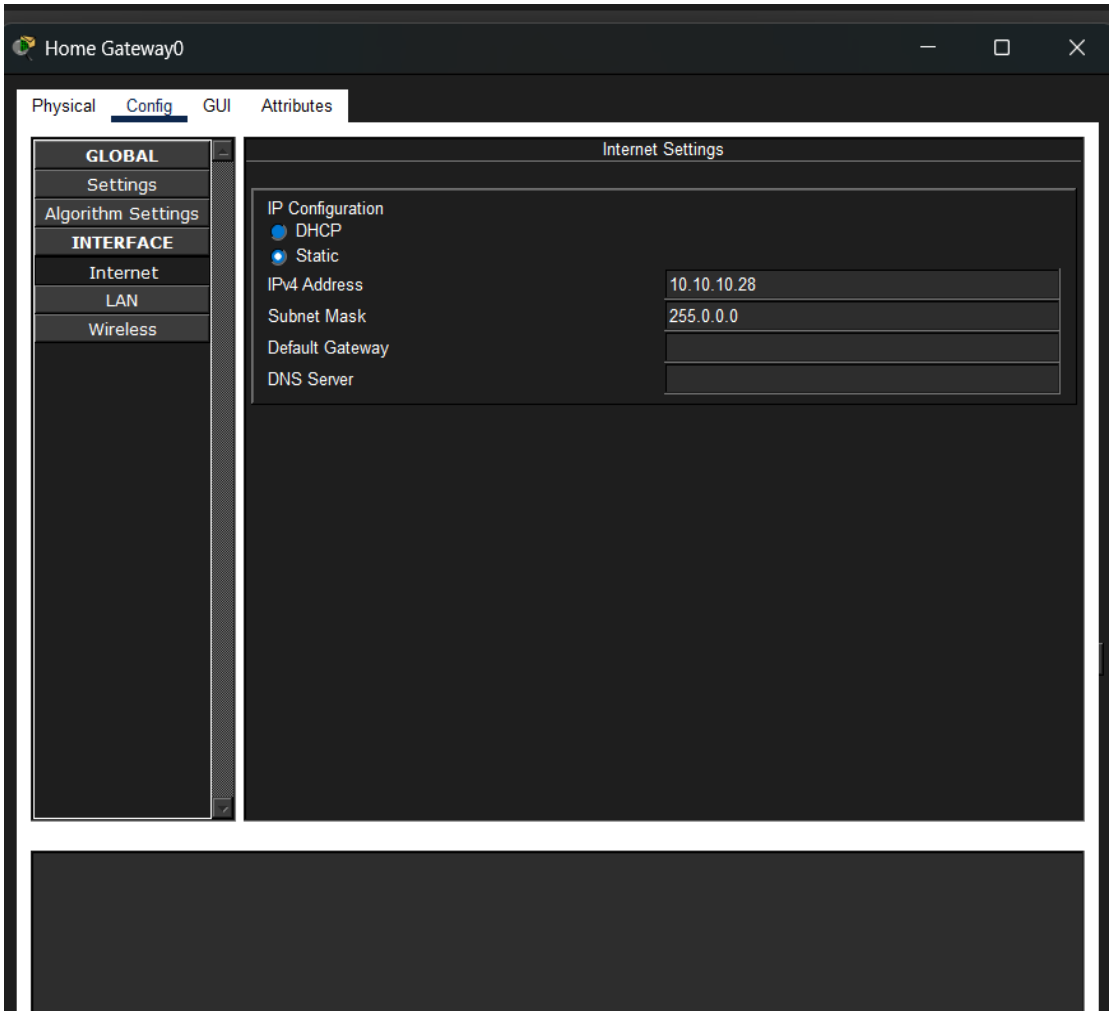
    - Group PCs into clusters (as shown: MYDESKTOP group, PC5–PC9 group, PC10–PC14 group, PC15–PC20 group)
    - Each group is connected to its own switch
3. Connect all switches to a **central device (Home Gateway0)** forming a campus backbone network.
4. Connect the **Server** to the central device for shared services.
5. Configure IP Addresses:
6. Configure routing in gateway if using multiple subnets.
7. Verify connectivity:
    - Use ping between PCs in different departments
    - Check communication with server

IP Configuration :
Dept1 PCs: 192.168.1.1 – 192.168.1.5
Dept2 PCs: 192.168.2.1 – 192.168.2.5
Dept3 PCs: 192.168.3.1 – 192.168.3.5
Dept4 PCs: 192.168.4.1 – 192.168.4.6

Result:
All departments successfully communicated through the central gateway. The server was accessible from all nodes, demonstrating proper campus-wide connectivity.

B. Static IP Assignment to Devices in Network



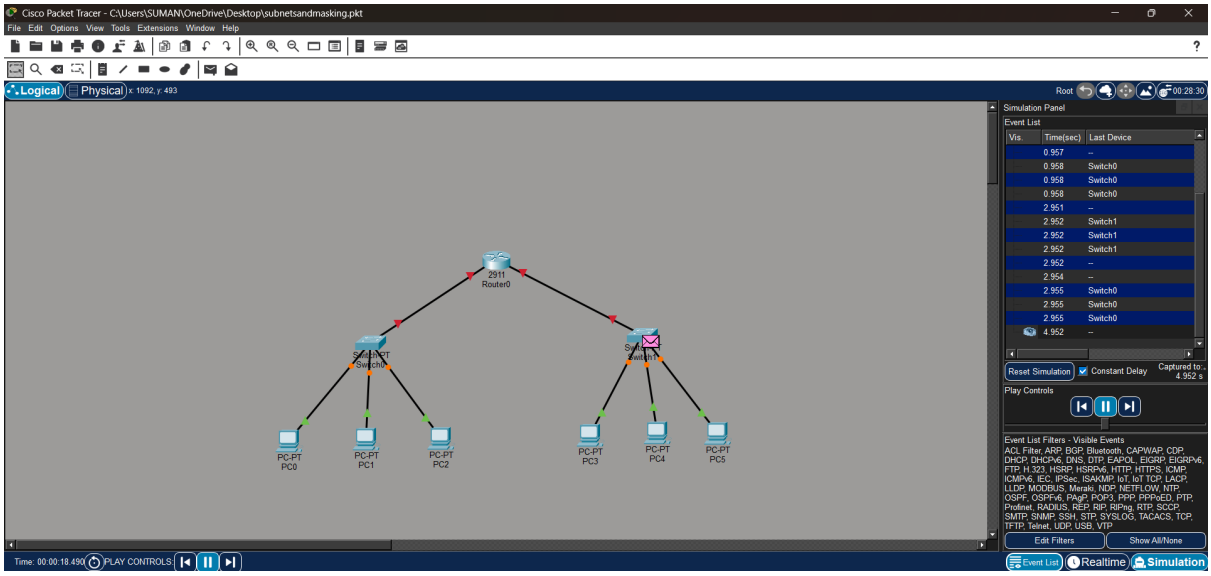


Static IP addressing was configured by manually assigning IP addresses to each device in the network using Cisco Packet Tracer.

This ensured that every device had a fixed and unique address, allowing stable and reliable communication. Connectivity was successfully verified using ping, confirming correct configuration.

EXPERIMENT6

IP and Subnetting-IP Configuration



Network Implementation

Aim:

To configure IP addresses using subnetting and enable communication between different networks.

Requirements:

- 1 Router (2911)
- 2 Switches
- 6 PCs (PC0–PC5)
- Cisco Packet Tracer

Procedure:

1. Create two LANs connected via a router.
2. Connect 3 PCs to each switch.
3. Connect both switches to the router.
4. Assign different subnets to each LAN.

5. Configure router interfaces for both networks.

IP Configuration:

LAN 1 (Left Side):
PC0: 192.168.1.1 / 255.255.255.0
PC1: 192.168.1.2 / 255.255.255.0
PC2: 192.168.1.3 / 255.255.255.0
Gateway: 192.168.1.254

LAN 2 (Right Side):
PC3: 192.168.2.1 / 255.255.255.0
PC4: 192.168.2.2 / 255.255.255.0
PC5: 192.168.2.3 / 255.255.255.0
Gateway: 192.168.2.254

Router Configuration:

- Interface 1: 192.168.1.254
- Interface 2: 192.168.2.254

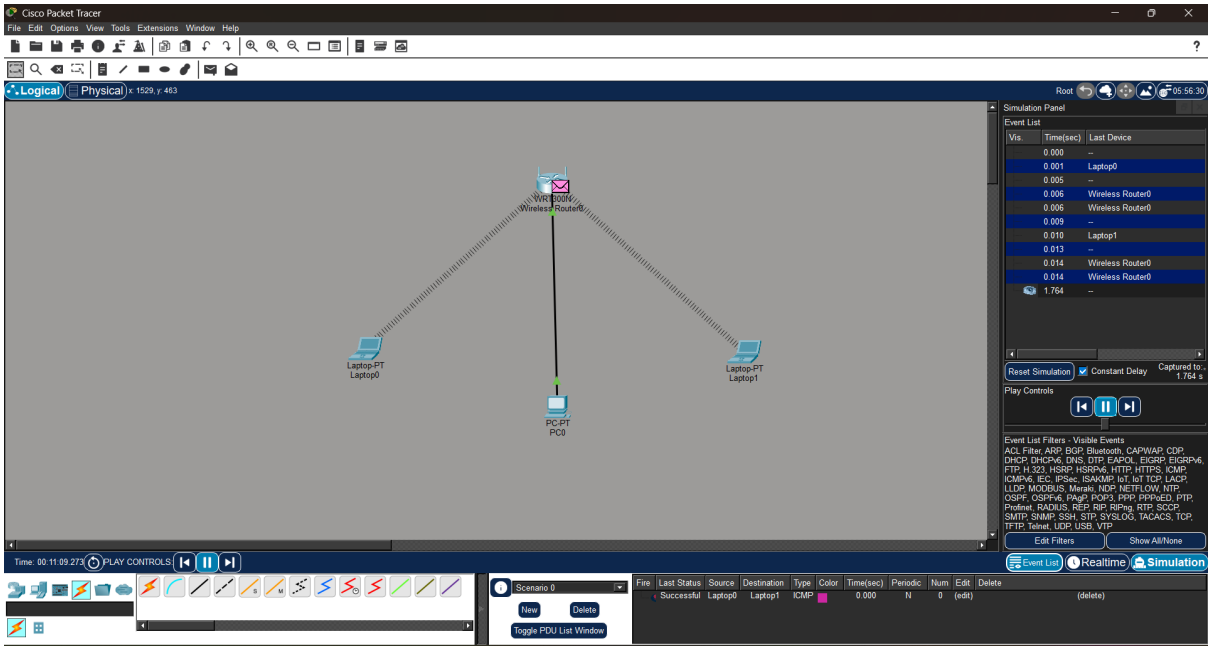
Subnetting:

- Network 1: 192.168.1.0 /24
- Network 2: 192.168.2.0 /24

Result:
Devices within the same subnet communicated directly, and communication between different subnets was successful through the router.

EXPERIMENT7

Wireless Home Network-Router configuration



Network Implementation

Aim:

To configure a wireless router and establish a wireless home network for communication between devices.

Requirements:

- 1 Wireless Router (WRT300N)
- 2 Laptops (Laptop0, Laptop1)
- 1 PC (PC0)
- Cisco Packet Tracer

Procedure:

1. Place the wireless router at the center of the network.
2. Add devices: Laptop0, Laptop1 (wireless) and PC0 (wired).
3. Connect PC0 to the router using a **copper straight-through cable**.

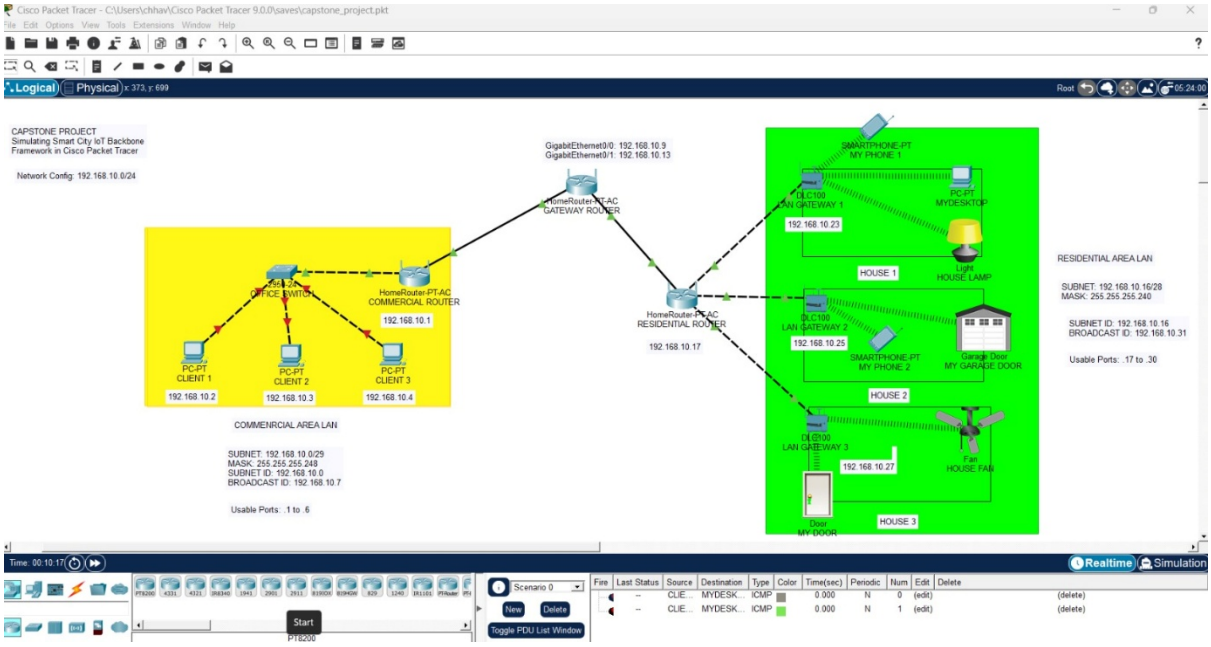
4. Configure the wireless router:
    - Go to **GUI / Config tab**
    - Set **SSID** (e.g., Home_Network)
    - Enable **WPA2-PSK security**
    - Set a password (e.g., 12345678)
5. Enable **DHCP service** on the router so that IP addresses are assigned automatically.
6. Connect laptops to Wi-Fi:
    - Go to Laptop → Desktop → PC Wireless
    - Select the SSID (Home_Network)
    - Enter the password
7. Verify connectivity:
    - Use ping between Laptop0 and Laptop1
    - Check communication between PC0 and laptops

IP Configuration:
All devices receive IP addresses automatically from the router (DHCP), typically in the range:
192.168.0.1/ 255.255.255.0

Result:
All devices (wired and wireless) successfully connected to the network. Laptops communicated wirelessly, and PC communicated through the router.

EXPERIMENT8

Capstone Project-Smart City Network



IP Configuration:

Backbone Network:

- Network: **192.168.10.0/24**
- Multilayer Switch / Backbone Router Interfaces:
  - G0/0: **192.168.10.1**
  - G0/1: **192.168.10.2**

Commercial Area LAN (Yellow Section):

- Network: **192.168.10.0/29**
- Gateway (Commercial Router): **192.168.10.1**
- PCs:
  - Client 1: **192.168.10.2**
  - Client 2: **192.168.10.3**
  - Client 3: **192.168.10.4**

Residential Area LAN (Green Section):

- Network: **192.168.10.16/28**
- Gateway (Residential Router): **192.168.10.17**

Devices:

- Smartphone 1: **192.168.10.18**
- Smartphone 2: **192.168.10.19**
- Laptop: **192.168.10.20**
- House Lamp (IoT): **192.168.10.21**
- Smart Door: **192.168.10.22**
- CCTV Camera: **192.168.10.23**

Devices Used:

- Multilayer Switch (Backbone)
- 2 Routers (Commercial + Residential)
- 2–3 Switches
- PCs (Clients)
- Smartphones & Laptop
- IoT Devices (Lamp, Door, Camera)
- Home Gateway

Types of Cables Used:

- **Copper Straight-Through Cable**
  - PC ↔ Switch
  - Switch ↔ Router
- **Copper Crossover Cable**
  - Switch ↔ Switch / Router ↔ Router (if directly connected)

- **Fiber Optic Cable**
    - Backbone (Multilayer Switch ↔ Routers)
- **Wireless (Wi-Fi)**
    - Smartphone, IoT devices connect via Home Gateway

Working of Network:

The network is divided into **two main areas: Commercial and Residential**. Both are connected through a **central multilayer backbone switch**, which manages routing between networks.

- Commercial area handles office/client communication
- Residential area connects IoT devices and smart home systems
- Home Gateway enables wireless connectivity for smart devices
- Backbone ensures communication between both networks

Estimated Cost:

Component	**Quantity**	**Cost (₹ approx)**
Multilayer Switch	1	25,000
Routers	2	10,000
Switches	2	6,000
PCs	3	75,000
IoT Devices	6–8	15,000
Cables & Setup	—	5,000
Total	—	**~1,30,000**

Advantages:

- Efficient separation of commercial and residential networks
- Smart home automation (IoT devices)
- Secure and scalable network design
- Centralized control using backbone network

Result:

The Smart City Network was successfully implemented using subnetting and proper IP addressing. Communication between commercial and residential areas was verified, and IoT devices functioned correctly within the network.